

A Minimal Model of the Singularity as Information Saturation

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Abstract

Classical general relativity predicts singularities where curvature diverges. Modern quantum-gravity models instead replace these divergences with finite transition points. This paper presents a minimal interpretation of the singularity as the moment when a collapsing spacetime reaches its maximum information density. At this point all degrees of freedom have been reduced to their minimal binary units, saturating the holographic entropy bound. Further compression is impossible, and the interior undergoes a bounce into a new spacetime region.

1. Idea

A collapsing black hole compresses matter and fields. As the radius decreases, the information density increases. The holographic principle limits the total information to

$$I_{\max} = \frac{A_{\text{app}}}{4L_P^2},$$

where A_{app} is the area of the *apparent* (dynamical) horizon. This horizon is quasi-local and evolves with infalling matter, so the interior is constrained by a local, not global, entropy bound.

The quantity I denotes the coarse-grained information content of the collapsing interior, defined as the effective number of independent degrees of freedom remaining after gravitational collapse erases longitudinal structure. Both I and $I_{\max}$ scale with the transverse area, ensuring a consistent comparison.

2. Singularity as a Limit State

The singularity is redefined as the moment when

$$I = I_{\max}.$$

At this stage all physical degrees of freedom have been reduced to binary form (0/1). This is a finite, physical limit rather than a divergence.

Because collapse eliminates one dimension of freedom, the remaining independent degrees of freedom scale as

$$I \propto r^2,$$

matching the holographic capacity

$$I_{\max} \propto A_{\text{app}} \propto r^2.$$

Thus the saturation condition is naturally reached at the minimum radius.

3. Bounce

At the minimum radius r_0 , the apparent-horizon area scales as $A_{\text{app}} \propto r_0^2$, fixing the holographic capacity $I_{\max} \propto r_0^2$, while the interior density scales as $\rho \propto M/r_0^3$. Because gravitational collapse eliminates longitudinal structure, the remaining independent degrees of freedom scale as $I \propto r_0^2$, so that at $r = r_0$ the interior information content exactly saturates the holographic bound ($I = I_{\max}$) at the same moment the critical density $\rho = \rho_c$ is reached.

The modified Friedmann equation,

$$H^2 = \alpha \rho \left(1 - \frac{\rho}{\rho_c} \right),$$

forces $H \rightarrow 0$ at $\rho = \rho_c$. Further collapse would require exceeding the holographic entropy bound, which is forbidden. Quantum-gravity corrections therefore reverse the sign of the effective gravitational dynamics, producing a bounce in which the interior transitions from contraction to expansion. The bounce is thus the dynamical response of spacetime to information saturation.

4. Summary

The singularity is reinterpreted as an information-saturation point: the final stage of collapse where all structure has been reduced to binary units and no further compression is possible. This provides a simple, non-singular description of black-hole interiors and aligns naturally with quantum-gravity bounce models.

Author Note

Paul Jarvis is an independent researcher based in the United Kingdom. He originally trained in electronics and applied physics, later expanding into theoretical physics through extensive self-directed study of established literature and contemporary research. His work focuses on conceptual models of quantum gravity and cosmology.